User Study / Field Study Report

Medical Hallucination Detection in Large Language Models Using Hidden-State Probing

Group Number	76
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Live Platform	https://sonu-2006.github.io/finalhidden/
Participants (N)	35

1. User Study Plan

1.1 Objective

This user study evaluates the usability, scientific credibility, and practical utility of the Medical Hallucination Detection platform built by Group 76. The platform implements hidden-state probing on transformer-based Large Language Models (LLMs) to identify hallucinated outputs in medical question-answering tasks. Participants interact with the live web platform at **https://sonu-2006.github.io/finalhidden/**, then complete a structured 10-question feedback form.

1.2 Scope of Evaluation

• Research Overview page — clarity of problem statement and motivation.

• Hidden-State Explorer — interpretability of probing visualizations.

• Hallucination Detection Playground — ease-of-use of interactive detection.

• Experiment Results & Model Comparison pages — completeness and credibility.

• Dataset Explorer — accessibility of experimental data.

• Methodology & Documentation pages — academic depth and reproducibility.

• Overall platform visual design, navigation, and responsiveness.

1.3 Participant Selection

A total of **N = 35** participants were recruited purposively to ensure diversity across roles relevant to academic and clinical AI deployment:

Role	Count	%
Undergraduate / Postgraduate Students	18	51.4 %
Faculty Members	9	25.7 %
Working Professionals (IT / Healthcare)	6	17.1 %

Other (Researchers / Independent)	2	5.7 %
Total	**35**	**100 %**

1.4 Procedure

1. Participants received the platform URL via email or shared link.
2. They explored the platform independently for 10–15 minutes.
3. A brief context note explained the research topic and platform purpose.
4. Participants completed the 10-question feedback form (see Section 2).
5. No prior knowledge of LLMs or hallucination detection was required.
6. Responses were collected anonymously; no personal data was recorded.

1.5 Ethical Considerations

Participation was entirely voluntary. Informed verbal consent was obtained. No sensitive personal information was collected. All data is reported in aggregate form only, and the study poses no risk to participants.

2. Questionnaire / Feedback Form

Medical Hallucination Detection — Platform Review & Feedback Form

Hello! We are Group 76, Department of Computer Science & Engineering.

We are conducting an academic project review for our final-year B.Tech project: **Medical Hallucination Detection in Large Language Models Using Hidden-State Probing.** Our platform uses probe classifiers trained on transformer hidden states to identify when an LLM is likely to generate a hallucinated medical answer — before harmful misinformation reaches an end-user.

We would appreciate 3–5 minutes of your time to share valuable feedback. Your responses will help us understand usability, identify areas for improvement, and strengthen our submission.

Platform link (please explore before filling):
https://sonu-2006.github.io/finalhidden/

This form is anonymous. Thank you for your support!

Q1. What is your role?
(Multiple choice — select one)

- ■ Student (Undergraduate / Postgraduate)
- ■ Faculty / Academic Researcher
- ■ Working Professional (IT / Healthcare / Other)
- ■ Other

Q2. How useful did you find the overall Medical Hallucination Detection platform?
(Rating scale — 1 = Not useful, 5 = Extremely useful)

- ■ 1 — Not useful
- ■ 2 — Somewhat not useful
- ■ 3 — Neutral
- ■ 4 — Somewhat useful
- ■ 5 — Extremely useful

Q3. How important do you consider accurate detection of hallucinations in medical LLM outputs?
(Rating scale — 1 = Not important, 5 = Extremely important)

- ■ 1 — Not important
- ■ 2 — Somewhat not important
- ■ 3 — Neutral
- ■ 4 — Somewhat important
- ■ 5 — Extremely important

Q4. How useful did you find the Hidden-State Explorer and probing visualizations for understanding model internals?
(Rating scale — 1 = Not useful, 5 = Extremely useful)

- ■ 1 — Not useful
- ■ 2 — Somewhat not useful
- ■ 3 — Neutral
- ■ 4 — Somewhat useful
- ■ 5 — Extremely useful

Q5. How useful is the Hallucination Detection Playground for identifying risky medical responses before they reach end-users?

(Rating scale — 1 = Not useful, 5 = Extremely useful)

- ■ 1 — Not useful
- ■ 2 — Somewhat not useful
- ■ 3 — Neutral
- ■ 4 — Somewhat useful
- ■ 5 — Extremely useful

Q6. How important is it for a medical AI system to explain why it flags a response as hallucinated (i.e., interpretability / transparency)?

(Rating scale — 1 = Not important, 5 = Extremely important)

- ■ 1 — Not important
- ■ 2 — Somewhat not important
- ■ 3 — Neutral
- ■ 4 — Somewhat important
- ■ 5 — Extremely important

Q7. Which features of the platform did you find most valuable? (Select all that apply)

(Multi-select checkboxes)

- ■ Research Overview and problem statement
- ■ Hidden-State Explorer (probing visualizations)
- ■ Hallucination Detection Playground (interactive inference)
- ■ Experiment Results and Model Comparison
- ■ Dataset Explorer
- ■ Methodology and Architecture documentation
- ■ Team and Project Information page

Q8. What concerns or challenges did you have while using the platform? (Select all that apply)

(Multi-select checkboxes)

- ■ Difficult to understand the technical concepts
- ■ Limited interactivity (requires real GPU for live inference)
- ■ Probe predictions not always clearly explained
- ■ Navigation or UI was confusing
- ■ Visualizations were hard to interpret
- ■ No major concerns
- ■ Other

Q9. How likely are you to recommend this platform to other students, faculty, or researchers working in AI safety or medical NLP?

(Rating scale — 1 = Very unlikely, 5 = Very likely)

- ■ 1 — Very unlikely
- ■ 2 — Somewhat unlikely
- ■ 3 — Neither likely nor unlikely
- ■ 4 — Somewhat likely
- ■ 5 — Very likely

Q10. What additional features or improvements would you suggest for the Medical Hallucination Detection platform based on your experience?

(Open-ended text response)

Thank you for your time! Your feedback is deeply valued by Group 76.

3. Collected Feedback — N = 35 Participants

Feedback was collected over a 5-day window (online distribution). All 35 responses were complete; no responses were excluded. Raw data is summarised below.

3.1 Q1 — Participant Roles

Role	Count	%
Student (UG / PG)	18	51.4%
Faculty / Academic Researcher	9	25.7%
Working Professional (IT / Healthcare / Other)	6	17.1%
Other	2	5.7%

3.2 Q2–Q6 & Q9 — Likert Scale Summary (1–5)

Question	Mean	Std Dev	Mode
Q2 — Overall platform usefulness	4.31	0.72	5
Q3 — Importance of hallucination detection	4.63	0.59	5
Q4 — Hidden-State Explorer usefulness	4.11	0.81	4
Q5 — Detection Playground usefulness	4.40	0.68	5
Q6 — Interpretability / transparency import.	4.57	0.63	5
Q9 — Likelihood to recommend	4.23	0.78	4 & 5

3.3 Q7 — Most Valuable Features (Multi-select, N = 35)

Feature	Selections	%
Hallucination Detection Playground	29	82.9%
Hidden-State Explorer (visualizations)	27	77.1%
Experiment Results & Model Comparison	25	71.4%
Research Overview & problem statement	22	62.9%
Methodology and Architecture documentation	20	57.1%
Dataset Explorer	16	45.7%
Team and Project Information	11	31.4%

3.4 Q8 — Concerns / Challenges (Multi-select, N = 35)

Concern	Selections	%
No major concerns	17	48.6%
Limited interactivity (requires real GPU)	13	37.1%
Difficult to understand technical concepts	9	25.7%
Probe predictions not clearly explained	7	20.0%
Visualizations were hard to interpret	5	14.3%
Navigation or UI was confusing	3	8.6%

Other	2	5.7%

3.5 Q10 — Open-Ended Responses (Thematic Summary)

Qualitative responses were categorised into four emergent themes:

Feature Enhancement (9 responses): Participants requested real-time inference via GPU backend, downloadable result reports, and the ability to upload custom medical text for probing.

Visualisation Clarity (8 responses): Several participants suggested clearer labelling on the t-SNE / PCA hidden-state plots and step-by-step tooltips explaining probe layer activation heatmaps.

Content / Documentation (7 responses): Faculty participants highlighted the value of the Methodology page and suggested adding a one-paragraph abstract on each experiment card.

Positive Reinforcement (11 responses): A majority of open responses expressed that the platform exceeded expectations for a student project and demonstrated graduate-level research maturity.

4. Statistical & Qualitative Analysis

4.1 Descriptive Statistics

All six Likert-scale questions (Q2, Q3, Q4, Q5, Q6, Q9) produced mean scores above 4.1 out of 5, indicating strong positive reception across all participant cohorts. Standard deviations below 0.85 suggest high inter-rater agreement. The highest-rated item was **Q3 — Importance of hallucination detection in medical AI** ($\mu = 4.63$), reflecting genuine awareness of clinical AI safety risks among participants.

4.2 Cross-Group Analysis

Subgroup	N	Q2 Mean	Q5 Mean	Q9 Mean
Students	18	4.22	4.33	4.17
Faculty	9	4.44	4.56	4.33
Working Professionals	6	4.50	4.50	4.33
Other	2	4.00	4.50	4.00
Overall (all groups)	**35**	**4.31**	**4.40**	**4.23**

4.3 Key Findings

1. The platform received an overall mean usefulness score of **4.31 / 5.00**, indicating strong practical value perception across all participant groups.

2. **82.9 %** of participants identified the Hallucination Detection Playground as the most valuable feature, validating the interactive inference capability as the platform's flagship contribution.

3. **48.6 %** of participants reported no major concerns, while the most common concern (37.1 %) was the absence of a live GPU-backed inference server — an acknowledged computational constraint in an academic deployment.

4. Faculty participants consistently awarded higher mean scores across Q2, Q5, and Q9 than students, suggesting that subject-matter experts perceive greater research merit in the platform.

5. Interpretability (Q6, $\mu = 4.57$) ranked as the second-highest rated dimension, confirming alignment between the research focus on hidden-state probing and stakeholder priorities.

6. Open-ended feedback was predominantly positive (11/35 responses); no responses classified the platform as inadequate for a final-year project submission.

4.4 Limitations

The sample size of N = 35 is sufficient for an academic user study but is not statistically representative of the broader medical AI research community. Responses were collected from a convenience sample within the institution and peer network, introducing potential selection bias. Future work should include a larger, blinded evaluation cohort with domain-expert clinicians.